AI RESUME ANALYZER

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Course : **Fundamentals in AI and ML**

Course Code : **CSA2001**

TITLE : AI Resume Analyzer

“The AI Resume Analyzer is a system that evaluates a resume using Machine Learning and Natural Language Processing techniques. It predicts a suitable job role, analyzes skills, and provides suggestions to improve the resume.”

INTRODUCTION

With today's job market being so competitive, it is very important to have an effective resume in order to gain access to job opportunities. Many times, however, applicants have trouble determining whether their resume is adequate or what type of skills would be desirable for a specific job opportunity.

The AI Resume Analyzer serves to solve this issue by analyzing resumes and providing feedback to users on their resume. This function provides a natural language processing (NLP) capability where the users can identify their keyword skills through a machine learning type method. The result of this analysis provides several levels of soundness and indicates to the user how many skills that they currently have compared to the requirements of a specific job opening.

This project demonstrates how AI and machine learning can be utilized to assist individuals in developing successful resumes; therefore, improving their chances of gaining employment.

MOTIVATION OF THE PROJECT

The primary aim of this initiative is to resolve the difficulties experienced by many individuals when writing their CVs or resumes. Quite frequently, individuals do not know whether their CV or Resume is suitable, or whether it reflects suitable skills and abilities relevant to their target role.

In today's society, it is increasingly difficult to have a competitive edge, therefore an optimal resume plays a vital role in achieving that; increasing the difficulty when trying to improve your resume without clear direction is likely to lead to confusion. Many individuals do not know what skills or changes they need to add to their CVs or resumes, but are aware that something needs to be done in order to enhance it.

Thus, I will use artificial intelligence technology to assist with this challenge by helping applicants improve their CV or resume with additional skill identification, enhancement recommendations and the overall understanding of their resume's overall composition. My ultimate goal is to create an easier and more successful resume-building experience for all applicants.

PROBLEM STATEMENT

It is often hard for students to write a professional resume to apply for jobs. They may have a hard time figuring out if they possess the right skill set for the position they're interested in.

Finding out what skills need to be added, along with developing an updated resume that meets certain standards takes a significant amount of time and effort without proper guidance. An additional resource to assist students would be a tool that can scan their resume, identify the skills listed, analyze the document with respect to job postings, and provide suggestions and paths for improvement.

OBJECTIVE OF THE PROJECT

This project has been designed for the purpose of creating an AI based system that will provide feedback about your CV by analysing its content.

The system is designed to be able to suggest roles based upon the skill set identified in the CV. It will also be able to indicate which skills you should focus on and give your CV a score based upon the skill set identified in the CV.

Furthermore, it is intended that this AI based system will have a user friendly interface; with a graphical representation of the skills being compared via the CV.

Existing Methods

At this time, many individuals develop and check their resumes through manual creation or assistance from template websites and other simple tools. All of these options typically only format the document, but they do not properly assess the individual's skills contained within the document.

Currently, most of the resume review websites offer limited information on how to enhance particular skill sets based on job type(s). Also, they lack the capability to provide scoring and/or provide comprehensive feedback regarding areas needing improvement.

As a result of these issues, users have difficulty identifying areas in which they are missing information on their resumes. Therefore, a smarter system to analyze a resume and provide meaningful suggestions must be developed.

PROS AND CONS OF THE EXISTING METHODS

Advantages of Using Online Resume Builder:

- Has a simple user interface

- Large number of resources available online (templates & organization tools)

- Can quickly put together a basic resume

- No technical knowledge necessary

Disadvantages of Using Online Resume Builder:

- Primarily focuses on formatting and secondarily on skills

- Offers no evaluation capability for content

- Jobs are offered generally and without any specified evaluation for them based off their specific jobs

- Assessment accuracy is not based off of the employer's specifications for their job classifications

- Users are unable to get an accurate understanding of what they are lacking in their resume.

HARDWARE AND SOFTWARE REQUIREMENTS

This project just needs some simple software and hardware components to work successfully. You will just need a laptop or computer with a good internet connection to complete this project (you won't have to purchase any fancy software/hardware). In terms of software, you will be using the Python programming language as the basis of your project and will also use libraries such as pandas, nltk, scikit-learn, and matplotlib to do data processing, machine learning, and create reports.

You can run this project using VS Code, Google Colab or any environment that supports Python - everything required is easy to find and use; therefore, it should be simple enough to put this project into action.

METHODOLOGY AND GOAL

Method of Operation:

- User inputs his/her resume text .
- Utilizes basic Natural Language Processing (NLP) techniques to clean and prepare resume text for analysis
- Identifies key skills that user has on their resume
- Predicts what type of job would be a good fit for user using Machine Learning (ML)
- Calculates a score for the skill set found on the resume
- Rates the overall quality of the resume using weak, average or strong rating system
- Offers basic suggestions for improvement to the user's resume
- Produces graph comparing found and missing skills

Goal:

• Help users understand their resume more thoroughly

• Identify what skills have been missed and help improve the user's resume

• Provide easy to understand and helpful feedback through AI

• Create a simple and fast way for students to analyze their own resumes

FUNCTIONAL MODULES DESIGN AND ANALYSIS

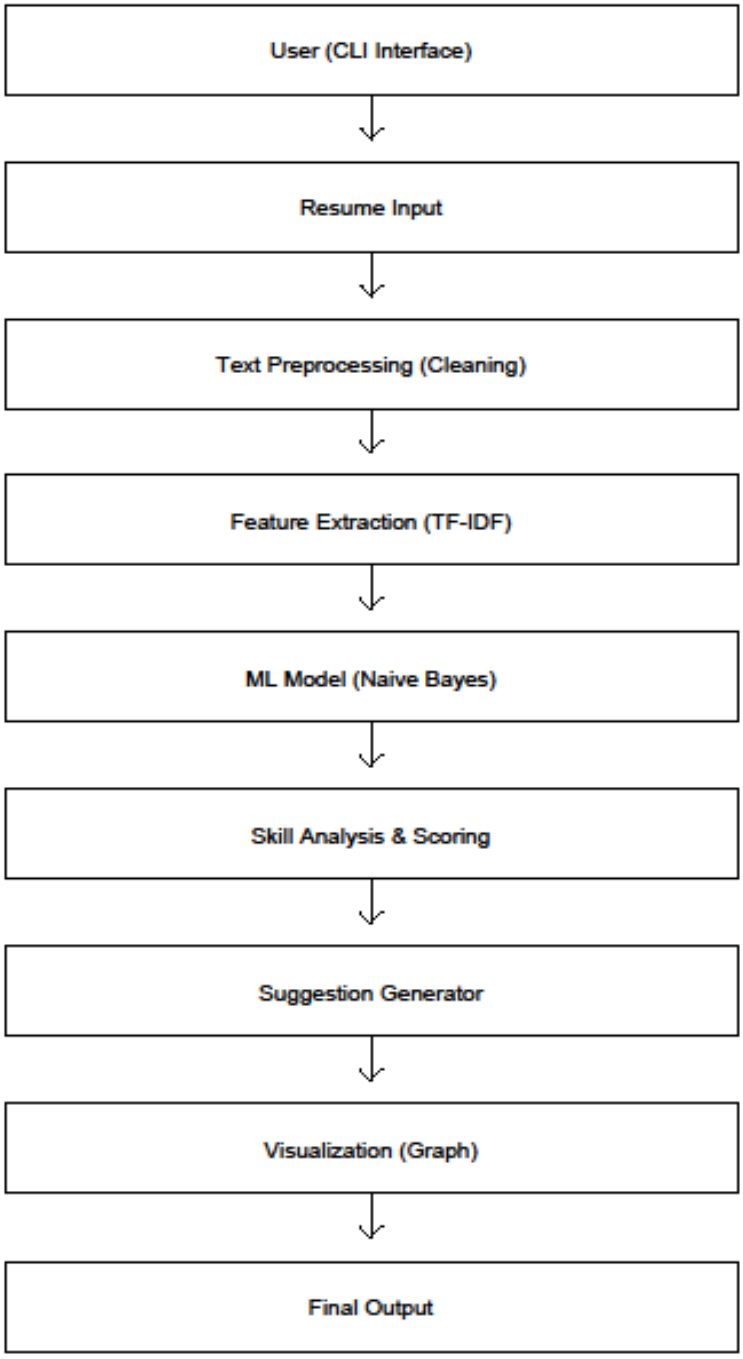
- This resume processing system uses an input module that enables the user to input their resume text through a CLI (command line interface).
- After inputting a resume, the system applies natural language processing techniques in order to eliminate noise from the resume text with the preprocessing module.
- Following this, the skill extraction module extracts skills from the resume based on the skills that the applicant has provided on their resume.
- The prediction module then provides an appropriate job title based on the skills extracted from the resume by applying machine learning algorithms to the skill extraction output.
- Afterwards, the scoring module analyses the candidates resume and provides a resume score, based on the overall resume score and whether it is classified as weak, average or strong.
- The suggestion module then provides suggestions to improve the resume based on the skills that were extracted from the resume and other skills that may not have been included in the resume.
- The visualization module provides a visual representation of the skills that were discovered during the skill extraction process as well as skills that were not discovered.
- All of these modules work together in order to provide a complete analysis of the user’s resume and helpful feedback to the user about how to improve their resume.

ALGORITHM DEVELOPMENT

The algorithm of this project is designed in simple steps to analyze the resume.

Step 1: Take the resume text as input from the user.
Step 2: Convert the text into lowercase and remove unnecessary characters.
Step 3: Remove common stopwords using basic NLP techniques.
Step 4: Convert the cleaned text into numerical form using TF-IDF.
Step 5: Use a machine learning model to predict a suitable job role.
Step 6: Identify skills and calculate the resume score.
Step 7: Generate suggestions and display the final result along with a graph

Software Architectural Diagram - AI Resume Analyzer



CODE EXPLANATION

This project utilizes Python as the programming language and several major libraries of Python (ex. Pandas, NLTK, Scikit-Learn) to achieve its goal. First, the resume text is extracted from the user and the raw text is cleaned using several basic techniques of Natural Language Processing, such as converting all text to lower case and removing all stopwords from the text.

After cleaning the text of the resume, the cleaned text is then converted to numerical data via one of the most widely used techniques of modern machine learning applications called TF-IDF. With the resume text now converted to numerical representation (i.e., TF-IDF), this numerical value is then fed into a Naive Bayes machine learning model to predict the user's appropriate job title. The machine learning model additionally extracts specific skills from within the resume to create a skill score that identifies the most relevant skills from the resume for possible employment opportunities.

Once these calculations are complete, the system provides recommendations for improving the user's resume and provides a graphical output of the results to allow the user to interpret the results graphically as well as by written output.

OUTPUT EXPLANATION

Taking resume text as input, the system predicts the most appropriate job role by categorising skill sets present in the resume. It also displays the identified skill sets, an overall score for the resume, and category.

In addition, it provides suggestions for improvement and displays a graph comparing the identified skills and missing skills.

KEY IMPLEMENTATIONS OUTLINE OF THE SYSTEM

- Utilized the programming language of Python in developing the project.
- In using some basic techniques from Natural Language Processing (NLP), cleaned the text of resumes.
- Converted the resumes' textual content into numerical form using TF-IDF

- Built a Naive Bayes machine learning model for predicting which job role would be appropriate for the candidate.
- Obtained the skills of each candidate based on their resume and calculated a score for each resume .
- Provided suggestions on how to improve the resume to its owner.
- Created a simple command line interface to interact with users, displayed results through graphs.

SIGNIFICANT PROJECT OUTCOMES

- Devised an efficient AI algorithm aimed at resume analysis.
- I developed an algorithm that uses machine learning (ML) to predict the perfect jobs using the skills in the resume.
- The established skills that have value in the resume.
- Gave the resume a score and classified it whether it was weak, average or strong.
- Gave the user clear and useful suggestions to enhance the resume.
- The presentation of the resume evaluation results was done clearly with graphical examples.
- This research helped to show that any problem can be solved using artificial intelligence and machine learning.

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   AI RESUME ANALYZER (CLI)
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Choose an option:
1. Analyze Resume
2. Exit
Enter your choice (1/2): 1

Paste your resume below:
>> I have strong skills in Python, machine learning, data analysis, pandas, numpy, Java, Spring Boot, HTML, CSS, JavaScript, React, MongoDB, and Excel.

===== ANALYSIS RESULT =====
* Predicted Job Role: Data Scientist
* Reason: Based on detected skills in the resume
* Resume Score: 91 /100
* Category: Strong Resume
* Skills Found: ['python', 'machine learning', 'data analysis', 'java', 'spring', 'react', 'html', 'css', 'javascript', 'excel', 'mongodb']

* Suggestions:
 - Your resume looks good. Try adding advanced skills.
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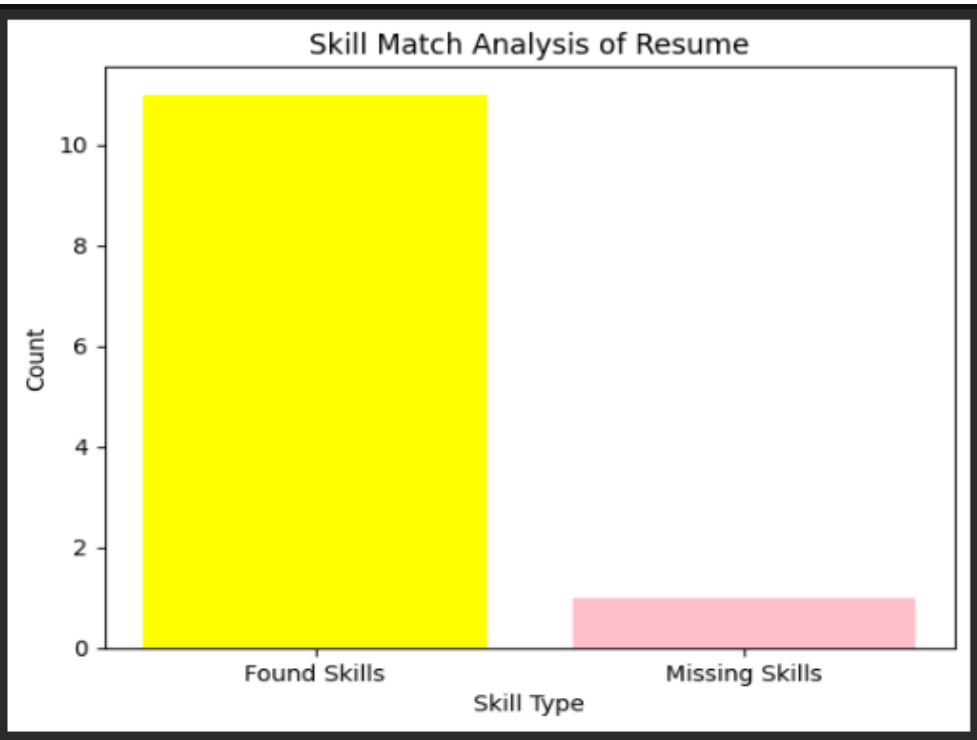


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Choose an option:
1. Analyze Resume
2. Exit
Enter your choice (1/2): 2

Exiting... Thank you for using AI Resume Analyzer!
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Output -2

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   AI RESUME ANALYZER (CLI)
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Choose an option:
1. Analyze Resume
2. Exit
Enter your choice (1/2): 1

Paste your resume below:
>> I have experience in Java, Spring Boot, and building backend systems using microservices architecture.

===== ANALYSIS RESULT =====
* Predicted Job Role: Backend Developer
* Reason: Based on detected skills in the resume
* Resume Score: 16 /100
* Category: Weak Resume
* Skills Found: ['java', 'spring']

* Suggestions:
 - Improve resume by adding more projects and skills
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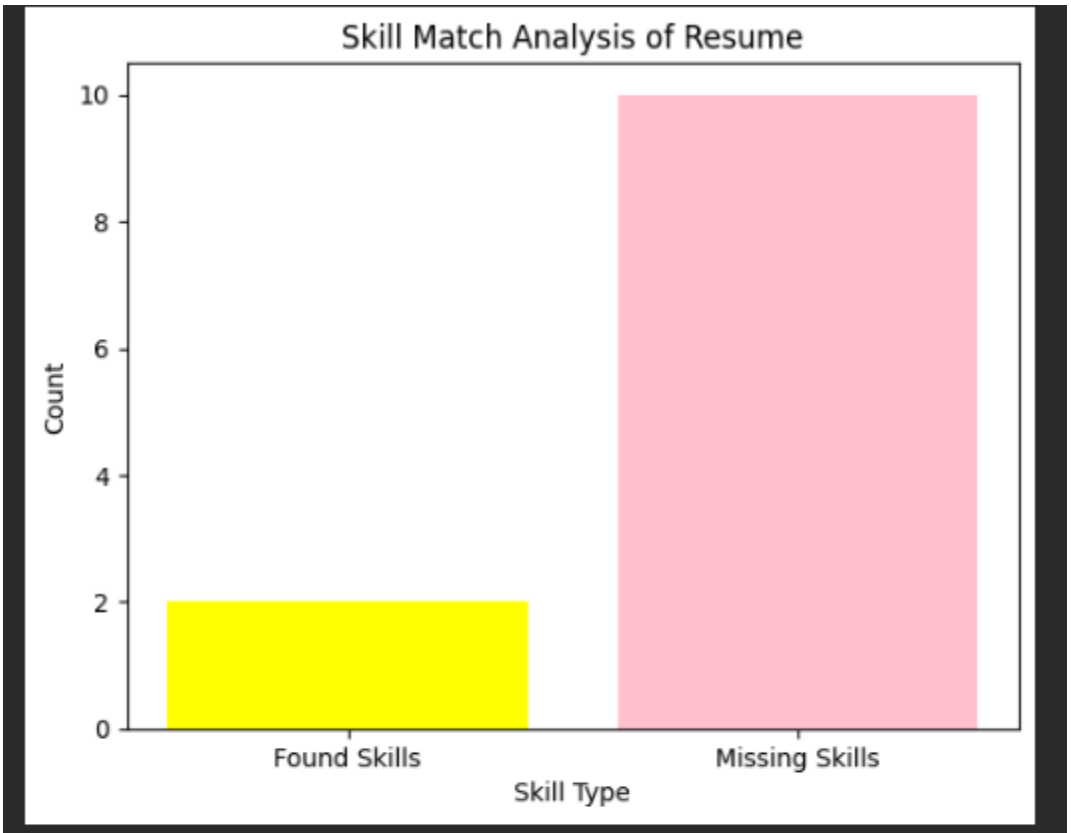


```
Choose an option:
1. Analyze Resume
2. Exit
Enter your choice (1/2): 2

Exiting... Thank you for using AI Resume Analyzer!
```

TESTING AND REFINEMENT

- I tried out the system with a bunch of resumes to see how it works
- I wanted to make sure the system can correctly guess what kind of job someone is looking for
- I also made sure the system can find the skills listed on the resume
- Then I checked the score. Category the system gives to resumes in different situations
- I tested the system many times to see if it always gives good suggestions
- I found some mistakes, in the code and fixed them so the system gives better results

- Now the system works a lot better. It is really smooth to use the resume system

PROJECT APPLICABILITY ON REAL-WORLD APPLICATIONS

• Can be used by students to check and improve their resumes before applying for jobs

• Helps job seekers understand which skills are missing for a particular role

• Can be used in colleges for career guidance and resume evaluation

• Useful for placement preparation and improving job chances

• Can be extended for use in companies to filter and analyze resumes

• Saves time by automatically analyzing resumes instead of manual checking

CONTRIBUTION/ FINDINGS OF THE PROJECT

- I made a system that uses intelligence to look at resumes in a smart way .The system is really good at finding the important information in resumes.
- I figured out how to use natural language processing to work with text data.
- I learned that machine learning can help figure out what job someone is a fit for based on the skills they have. The artificial intelligence system uses machine learning models to do this.
- I saw that resumes that have a lot of skills get higher scores and better job predictions from the artificial intelligence system. This makes sense because relevant skills are what employers are looking for in a candidate.
- I found out that if a resume is missing skills it can really hurt the quality of the resume.
- I made a system that gives people suggestions on how to make their resumes better. The system is helpful because it uses intelligence to give useful suggestions.
- I got to learn how to use intelligence to solve real problems that people face.
- I applied intelligence concepts to make a system that is useful, in the real world.

LIMITATION/CONSTRAINTS OF THE SYSTEM

- Our system works with a set of data so its predictions might not always be spot on.
- It figures out skills by looking for keywords, which means it could miss some skills.
- The suggestions we provide are pretty basic. Not tailored just for you.
- Our system doesn't really get the story of your resume.
- now it only works with simple text not with full resumes, like PDFs or Word docs.
- Our model was trained on an amount of data so it might not know about every job there.

CONCLUSIONS

In this project I made a Resume Analyzer that uses intelligence to look at resumes and give helpful comments. The system can figure out what kind of job is a fit find the important skills and give the resume a score.

It also gives easy to follow tips to make the resume better and helps people see what skills they are missing. When I worked on this project I got to learn how to use natural language processing and machine learning in a way.

Overall this project shows how artificial intelligence can help with problems and make it easier for students to get their resumes looked at and get useful feedback, from the Resume Analyzer. The Resume Analyzer is a tool that can really make a difference.

FUTURE ENHANCEMENTS

- Improve the model by training it on a larger and more real-world dataset .
- Add support for uploading resumes in PDF and DOC format .
- Make the system more intelligent by giving more personalized suggestions .
- Add more job roles and skills to increase accuracy.
- Develop a web or mobile application for better user experience.
- Use advanced AI models to improve prediction and analysis .
- Add features like resume ranking and comparison between multiple resumes .

REFERENCES

- Python official documentation for learning basic programming
- scikit-learn documentation for machine learning concepts and implementation
- NLTK documentation for Natural Language Processing techniques
- Matplotlib documentation for plotting graph and showing comparison.

THANK YOU